MTH 204 : Lecture 11

Differential Operator :

- Here an operator is a transformation that transforms a function into another function.

(Recall : Transformation from a vector space to another vector space)

- The Differential operator D transforms a differentiable function into its derivative.

e.g. consider $D : C^1 \longrightarrow C$

where C^1 is the space of all differentiable functions with continuous derivative

and C is the space of all continuous functions.

Notation: $D = \frac{d}{dx}$

So, $Dy = y' = \frac{dy}{dx}$

For higher derivatives $D^2 y = D(Dy) = y''$

For a homogeneous linear ODE with constant Coefficients $y'' + ay' + by = 0$ can be written as $Ly = P(D)y = 0$

where $L = P(D) = D^2 + aD + bI$

(second order differential operator)

Here I is the identity operator: $Iy = y$

Note that L is defined only on C^2.

- <u>L is a Linear operator:</u>

If y and w are two twice differentiable functions, then

$L(cy + dw) = cL(y) + dL(w)$ where c and d are constants.

$$L(e^{\lambda x}) = P(D)(e^{\lambda x}) = (D^2 + aD + bI)(e^{\lambda x})$$

$$= (\lambda^2 + a\lambda + b)(e^{\lambda x}) = P(\lambda)(e^{\lambda x}) = 0$$

Thus $e^{\lambda x}$ is a solution of the ODE iff λ is a solution of the characteristic equation $P(\lambda) = 0$

- $P(\lambda)$ is a polynomial in the usual sense of algebra. Replacing λ by the operator, we get the operator polynomial $P(D)$: Operational Calculus.

- $P(D)$ can be treated like an algebraic quantity and we can factor it.

Ex: Factor and solve:

$$\left(D^2 - 4.00D + 3.84I\right)y = 0$$

$$\Rightarrow \quad (D - 2.4I)(D - 1.6I)y = 0$$

So, $y_1 = e^{2.4x}$ and $y_2 = e^{1.6x}$ is a basis of the solution and the general solution is:

$$y = c_1 e^{2.4x} + c_2 e^{1.6x}$$

Note: If $L(D)z = 0$ then z is an eigen function of L corresponding to the eigenvalue 0.

Note:
Let $L(D)y = y$ i.e. y is an eigenfunction of L corresponding to eigen value 1.

Then $\quad (D^2 + aD + bI)y = y.$

$$\Rightarrow \left[D^2 + aD + (b-1)I\right]y = 0$$

$\Rightarrow y$ is eigenfunction of $L_1 = D^2 + aD + (b-1)I$ corresponding to the eigen value 0.

Thus solution of a Homogeneous Linear Differential Equation can be thought of eigen function of the corresponding Differential Operator.

<u>Ex:</u> Find all eigen functions corresponding to zero eigen value of the differential operator $D^2 - 2D + 3I$.

- $(D^2 - 2D + 3I)\, y = 0$

$$\Rightarrow y'' - 2y' + 3y = 0$$

Characteristic Equation: $\lambda^2 - 2\lambda + 3 = 0$

$$\Rightarrow \lambda = \frac{2 \pm \sqrt{4-12}}{2} = 1 \pm \sqrt{2}\, i$$

So, $y(x) = e^{x}\left[c_1 \cos(\sqrt{2}\,x) + c_2 \sin(\sqrt{2}\,x)\right]$

<u>Ex:</u> If we want the eigen function of $(D^2 - 2D + 3I)$ corresponding to eigenvalue 1, then $(D^2 - 2D + 3I)\, y(x) = y(x)$ and so we need to find the solution of

$$y'' - 2y' + 2y = 0$$

The characteristic Equation is $\lambda^2 - 2\lambda + 2 = 0$

$$\Rightarrow \lambda = \frac{2 \pm \sqrt{(-2)^2 - 4 \times 2}}{2} = \frac{2 \pm \sqrt{-4}}{2} = 1 \pm i$$

So, $y(x) = e^{x}\left[c_1 \cos(x) + c_2 \sin x\right]$

is an eigen function of $D^2 - 2D + 3I$

corresponding to to eigen value 1.

<u>Note:</u> Let $y'' + ay' + by = 0$

$$\Leftrightarrow (D^2 + aD + bI)y = 0$$

$$\Leftrightarrow L(D)y = 0 \text{ where } L(D) = D^2 + aD + bI$$

$$\Leftrightarrow y \in \ker L(D)$$

Thus, solving the ODE, $y'' + ay' + by = 0$

is same as finding kernel of $L = D^2 + aD + bI$

<u>Ex:</u> Find the kernel of $D^2 - 11D - 60I$

$$y \in \ker(D^2 - 11D - 60I)$$

$$\Leftrightarrow (D^2 - 11D - 60I)y = 0$$

Characteristic Equation: $\lambda^2 - 11\lambda - 60 = 0$

$$\Rightarrow (\lambda - 15)(\lambda + 4) = 0$$

$$\Rightarrow \lambda = 15, -4$$

Thus $\ker(D^2 - 11D - 60I)$

$$= \{c_1 e^{15x} + c_2 e^{-4x} : c_1, c_2 \in \mathbb{R}\}$$

Higher Order Linear ODEs

- A nth order ODE is linear if it can be written as:

$$y^{(n)}(x) + p_{n-1}(x)\, y^{(n-1)}(x) + \cdots + p_1(x) y'(x) + p_0(x) y(x) = r(x)$$

It is homogeneous if $r(x) = 0$.

Otherwise it is nonlinear.

Theorem (Principle of Superposition):

The linear combination of any number of solutions of a homogeneous linear ODE is also a solution.

General Solution:

The general solution of the ODE on an open interval I is of the form

$$y = c_1 y_1 + c_2 y_2 + \cdots + c_n y_n$$

where $\{y_1, y_2, \ldots, y_n\}$ is a set of linearly independent (in I) solutions.

- A set $\{y_1, y_2, \ldots, y_n\}$ is linearly indepent in I if

$$c_1 y_1 + c_2 y_2 + \cdots + c_n y_n = 0$$
$$\Rightarrow c_1 = c_2 = \cdots = c_n = 0$$

Ex: $\{x^2, 5x, 2x\}$ is linearly dependent

Since $0(x^2) + 1(5x) + \left(-\frac{5}{2}\right)(2x) = 0$

Ex: $\{1, x, x^2\}$ is linearly independent on any open interval

Since $c_1 \times 1 + c_2 x + c_3 x^2 = 0 = 0 \times 1 + 0 \times x + 0 \times x^2$

$$\Rightarrow c_1 = 0, \; c_2 = 0, \; c_3 = 0$$

- We will learn method of testing independence with the Wronskian.

Basis of a General solution:

Ex: $y^{(4)} - 29y'' + 100y = 0$

Let us try a solution of the form $e^{\lambda x}$

Then $\lambda^4 e^{\lambda x} - 29\lambda^2 e^{\lambda x} + 100e^{\lambda x} = 0$

$\Rightarrow \quad e^{\lambda x}(\lambda^4 - 29\lambda^2 + 100) = 0$

$\Rightarrow \quad \lambda^4 - 29\lambda^2 + 100 = 0$

$\Rightarrow \quad (\lambda^2 - 25)(\lambda^2 - 4) = 0 \Rightarrow \lambda^2 = 25$ or $\lambda^2 = 4$

$\Rightarrow \quad \lambda = 5, -5, 2, -2$

So, the general solution is:

$$y = c_1 e^{5x} + c_2 e^{-5x} + c_3 e^{2x} + c_4 e^{-2x}$$

Initial Value Problem:

An initial value problem consists of n initial conditions

$$y(x_0) = K_0, \quad y'(x_0) = K_1, \ldots, y^{(n-1)}(x_0) = K_{n-1}$$

These n conditions are used to determine the constants of the general solution:

$$y = c_1 y_1 + c_2 y_2 + \cdots + c_n y_n$$

Ex: $x^3 y''' + 2x^2 y'' - 4xy' + 4y = 0, \; y(1) = 2, \; y'(1) = 1, \; y''(1) = 4$

This is a third order Euler-Cauchy problem.

Let us try a solution $y = x^m$

Then $x^3\left[m(m-1)(m-2)x^{m-3}\right]+2x^2\left[m(m-1)x^{m-2}\right]$
$$-4x\left(mx^{m-1}\right)+4x^m=0$$

$$\Rightarrow \left[m(m-1)(m-2)+2m(m-1)-4m+4\right]x^m=0$$

$$\Rightarrow (m-1)\left[m^2-\cancel{2m}+\cancel{2m}-4\right]x^m=0$$

$$\Rightarrow (m-1)(m-2)(m+2)\,x^m=0$$

$\Rightarrow m=1,2,-2$ and the general solution is : $\boxed{y=c_1x+c_2x^2+c_3x^{-2}}$

For the particular solution, $y(1)=2,\ y'(1)=1,\ y''(1)=4$

Hence, $c_1+c_2+c_3=2$ .... (1)

$y'(x)=c_1+2c_2x-2c_3x^{-3} \Rightarrow c_1+2c_2-2c_3=1$ ..... (2)

$y''(x)=2c_2+6c_3x^{-4} \Rightarrow 2c_2+6c_3=4$ ...... (3)

Now (1) $-$ (2) $\Rightarrow -c_2+3c_3=1 \Rightarrow -2c_2+6c_3=2$

and (3) $\Rightarrow 2c_2+6c_3=4$

Adding $12c_3=6 \Rightarrow c_3=\boxed{\frac{1}{2}}$

Then (3) $\Rightarrow 2c_2+6\left(\frac{1}{2}\right)=4 \Rightarrow 2c_2=4-3 \Rightarrow c_2=\boxed{\frac{1}{2}}$

and (1) $\Rightarrow \boxed{c_1=1}$

So, the particular solution is :

$$\boxed{y_p=x+\frac{1}{2}x^2+\frac{1}{2}x^{-2}}$$

Linear Independence of Solutions : Wronskian :

Consider the homogeneous linear ODE :

$$y^{(n)}(x) + p_{n-1}(x)\, y^{(n-1)}(x) + \cdots\cdots + p_1(x)\, y'(x) + p_0(x)\, y(x) = 0$$

Assume that the functions $p_{n-1}(x), \ldots, p_1(x), p_0(x)$ are continuous on an open interval I.

A set of solution $\{y_1, y_2, \ldots, y_n\}$ on I is linearly dependent iff their Wronskian,

$$W(x) = W(y_1, \ldots, y_n) = \begin{vmatrix} y_1 & y_2 & \cdots & y_n \\ y_1' & y_2' & \cdots & y_n' \\ \vdots & & & \\ y_1^{(n-1)} & y_2^{(n-2)} & \cdots\cdots & y_n^{(n-1)} \end{vmatrix} = 0$$

for some $x_0 \in I$.

Furthermore, if $W(x) = 0$ for some $x = x_0 \in I$, then $W(x) = 0 \;\forall\; x \in I$

Hence if there is an $x_1 \in I$ at which W is not zero, then $y_1, y_2, \ldots, y_n$ are linearly independent on I, so that they form a basis of solutions of 2 on I.

Ex: $1, x, x^2$ are solutions of the ODE:

$$y''' = 0$$

The coefficient functions are continuous everywhere (i.e. on $\mathbb{R}$)

Now $W(x) = \begin{vmatrix} 1 & x & x^2 \\ 0 & 1 & 2x \\ 0 & 0 & 2 \end{vmatrix} = 1(2) = 2 \neq 0$

Hence $1, x, x^2$ are linearly independent on $\mathbb{R}$ and forms a basis of solutions of the ODE.

Ex: $x^2, 5x, 2x$ are solutions of the ODE

$$y''' = 0$$

However,

$W(x) = \begin{vmatrix} x^2 & 5x & 2x \\ 2x & 5 & 2 \\ 2 & 0 & 0 \end{vmatrix} = 2(10x - 10x)$

$= 0$

Hence $x^2, 5x, 2x$ are linearly dependent.

Existence and Uniqueness of Solutions:

Existence and Uniqueness Theorem for IVP:

Consider the Initial Value problem:

$$y^{(n)} + p_{n-1}(x)\, y^{(n-1)}(x) + \cdots\cdots + p_1(x)\, y'(x) + p_0(x)\, y(x) = 0$$

with n initial conditions:

$$y(x_0) = K_0,\ y'(x_0) = K_1, \ldots\ldots, y^{(n-1)}(x_0) = K_{n-1}.$$

If the coefficients $p_{n-1}(x), \ldots, p_1(x)$ are continuous on some open interval I and $x_0 \in I$, then the IVP has a unique solution $y(x)$ on I.

Existence of a General solution:

Consider the homogeneous linear ODE:

$$y^{(n)}(x) + p_{n-1}(x)\, y^{(n-1)}(x) + \cdots\cdots + p_1(x)\, y'(x) + p_0(x)\, y(x) = 0$$

If the coefficient functions $p_{n-1}(x), \ldots, p_1(x), p_0(x)$ are continuous on some open interval I, then there exists a general solution of the ODE on I.

General solution Includes All Solutions:

Consider the homogeneous linear ODE

$$y^{(n)}(x) + p_{n-1}(x)\, y^{(n-1)}(x) + \cdots + p_1(x)\, y'(x) + p_0(x)\, y(x) = 0$$

If the coefficient functions $p_{n-1}(x), \ldots, p_1(x), p_0(x)$ are continuous on some open interval I, then every solution $y = Y(x)$ on I is of the form: $Y(x) = c_1 y_1(x) + \cdots + c_n y_n(x)$ where $\{y_1, y_2, \ldots, y_n\}$ is a basis of solutions of ODE on I and $c_1, c_2, \ldots, c_n$ are suitable constants.

Hence the ODE has no singular solution (that is, solutions that cannot be obtained from the general solution)

Homogeneous Linear ODEs with constant coefficients:

Let us consider an n-th order homogeneous linear ODE with constant coefficients

$$y^{(n)} + p_{n-1}\, y^{(n-1)} + \cdots + p_1 y' + p_0 y = 0$$

If we substitute $y = e^{\lambda x}$ as a possible solution, we get the characteristic equation as:

$$\lambda^n + p_{n-1}\lambda^{n-1} + \cdots + p_1\lambda + p_0 = 0$$

where roots may be real or complex, and have any degree of multiplicity.

- <u>Distinct Real Roots:</u>

If all the n roots $\lambda_1, \lambda_2, \ldots, \lambda_n$ are real and different, then the corresponding general solution is $y = c_1 e^{\lambda_1 x} + \cdots + c_n e^{\lambda_n x}$

(The n solutions $e^{\lambda_1 x}, e^{\lambda_2 x}, \ldots, e^{\lambda_n x}$ are linearly independent).

- <u>Simple complex roots:</u>

The (part of the) general solution associated to a couple of conjugated (simple) complex roots $\lambda_1, \lambda_2 = \alpha \pm i\omega$ is

$$e^{\alpha x}\left(c_1 \cos(\omega x) + c_2 \sin(\omega x)\right) = e^{\alpha x}\cos(\omega x - \delta)$$

<u>Ex:</u> $\quad y''' - 5y'' + 289y' - 1445y = 0$

The characteristic equation is:

$$\lambda^3 - 5\lambda^2 + 289\lambda - 1445 = 0$$

$$\Rightarrow \quad \lambda^2(\lambda-5) + 289(\lambda-5) = 0$$

$$\Rightarrow (\lambda-5)(\lambda^2+289) = 0 \Rightarrow \lambda = 5, \pm 17i$$

So, the general solution is:

$$y = c_1 e^{5x} + c_2 \cos(17x) + c_3 \sin(17x)$$

Multiple Real roots:

The (part of the) general solution associated to a real root λ of order m is

$$\left(c_0 + c_1 x + \cdots + c_{m-1} x^{m-1}\right) e^{\lambda x}$$

Ex: $y^{(5)} - 12y^{(4)} + 48y''' - 64y'' = 0$

The characteristic equation is:

$$\lambda^5 - 12\lambda^4 + 48\lambda^3 - 64\lambda^2 = 0$$

$$\Rightarrow \quad \lambda^2(\lambda^3 - 12\lambda^2 + 48\lambda - 64) = 0$$

$$\Rightarrow (\lambda-0)^2(\lambda-4)^3 = 0$$

$$\Rightarrow \quad \lambda = 0 \text{ (multiplicity 2)}$$

and $\lambda = 4$ (multiplicity 3)

Hence the general solution is:

$$y = (c_1 + c_2 x + c_3 x^2) e^{4x} + (c_4 + c_5 x) e^{0.x}$$

$$\Rightarrow \quad y = (c_1 + c_2 x + c_3 x^2) e^{4x} + (c_4 + c_5 x)$$

Multiple Real roots : Proof :

Consider the Linear ODE with constant coefficients

$$y^{(n)} + p_{n-1} y^{(n-1)} + \cdots + p_1 y' + p_0 y = 0$$

and define the differential operator :

$$L = D^n + p_{n-1} D^{n-1} + \cdots + p_1 D + p_0 I$$

The ODE can be written as $Ly = 0$

For $y = e^{\lambda x}$ is a solution of the ODE then,

$$L(e^{\lambda x}) = e^{\lambda x} P(\lambda) = 0$$

where $P(\lambda) = \lambda^n + p_{n-1}\lambda^{n-1} + \cdots + p_1\lambda + p_0$

Now if λ_1 is a root of $P(\lambda)$ of order m, then

$$L(e^{\lambda x}) = e^{\lambda x}(\lambda-\lambda_1)^m P_1(\lambda)$$ where $P_1(\lambda)$ is a polynomial of degree $(n-m)$ and $P_1(\lambda_1) \neq 0$

$$\frac{\partial L(e^{\lambda x})}{\partial \lambda} = m(\lambda-\lambda_1)^{m-1}\left[e^{\lambda x} P_1(\lambda)\right] + (\lambda-\lambda_1)^m \frac{\partial\left[e^{\lambda x} P_1(\lambda)\right]}{\partial \lambda} \quad \ldots\ldots (1)$$

Since all derivatives are continuous,

$$\frac{\partial L(e^{\lambda x})}{\partial \lambda} = L\left(\frac{\partial (e^{\lambda x})}{\partial \lambda}\right) = L(x e^{\lambda x})$$

$$\Rightarrow \left.\frac{\partial L(e^{\lambda x})}{\partial \lambda}\right|_{\lambda=\lambda_1} = L(x e^{\lambda_1 x}) \quad \ldots\ldots (2)$$

From ① and ②

$$L(xe^{\lambda_1 x}) = \left.\frac{\partial L(e^{\lambda x})}{\partial \lambda}\right|_{\lambda=\lambda_1} = m(\lambda_1-\lambda_1)^{m-1}\left[e^{\lambda_1 x}P_1(\lambda_1)\right] + (\lambda_1-\lambda_1)^m \left.\frac{\partial\left[e^{\lambda x}P_1(\lambda)\right]}{\partial \lambda}\right|_{\lambda=\lambda_1}$$

$$= 0$$

$\Rightarrow \quad L(xe^{\lambda_1 x}) = 0$

$\Rightarrow \quad xe^{\lambda_1 x}$ is a solution of the ODE.

We can repeat this step and produce solutions $x^2e^{\lambda_1 x}, \ldots\ldots, x^{m-1}e^{\lambda_1 x}$ by another $(m-2)$ such differentiations with respect to λ. Going one step further will no longer give zero on the right because the lowest power of $\lambda-\lambda_1$ would then be $(\lambda-\lambda_1)^0$ and this will be multiplied by $m!\,P_1(\lambda)$ where $P_1(\lambda_1) \neq 0$ ($P_1(\lambda)$ has no factor $(\lambda-\lambda_1)$).

Thus $e^{\lambda_1 x}, \; xe^{\lambda_1 x}, \ldots\ldots, x^{m-1}e^{\lambda_1 x}$ are precisely the solutions associated to λ.

By calculating the Wronskian of the above solutions, we can see that it is $e^{\lambda m x}$ times a determinant which is the wronskian of $1, x, x^2, \ldots\ldots x^{m-1}$. The later is $(1!\,2!\ldots(m-1)!) \neq 0$

$(1, x, x^2, \ldots, x^{m-1}$ are solutions of $y^{(m)} = 0)$

Therefore $e^{\lambda_1 x}, xe^{\lambda_1 x}, \ldots\ldots, x^{m-1}e^{\lambda_1 x}$ are linearly independent solutions of the ODE.

Multiple Complex Conjugate roots:

The part of the general solution associated to a couple of complex conjugate roots

$$\lambda_1, \lambda_2 = \alpha \pm i\omega$$

of order m is :

$$e^{\alpha x}\Big(\left(c_0 + c_1 x + \cdots + c_{m-1}x^{m-1}\right)\cos(\omega x) + \left(d_0 + d_1 x + \cdots\cdots + d_{m-1}x^{m-1}\right)\sin(\omega x)\Big)$$

or can be written as :

$$e^{\alpha x}\left(A_0\cos(\omega x - \delta_0) + A_1 x\cos(\omega x - \delta_1) + \cdots\cdots + A_{m-1}x^{m-1}\cos(\omega x - \delta_{m-1})\right)$$

Nonhomogeneous Linear ODEs:

A nonhomogeneous linear ODEs of nth order is given by $y^{(n)} + p_{n-1}(x)\,y^{(n-1)} + \cdots\cdots + p_1(x)\,y = r(x)$

and $r(x) \not\equiv 0$

A general solution on an open interval I is of the form $y(x) = y_h(x) + y_p(x)$

where $y_h(x) = c_1 y_1(x) + \cdots + c_n y_n(x)$ is a general solution of the corresponding homogeneous ODE $y^{(n)} + p_{n-1}(x)\, y^{(n-1)} + \cdots + p_1(x)\, y' + p_0(x)\, y = 0$ on I.

y_p is any solution of the nonhomogeneous ODE on I containing no arbitrary constants.

If $p_{n-1}(x), \ldots, p_1(x)$ and $r(x)$ are continuous functions on I, then a general solution of the ODE exists and includes all solutions. Thus the ODE has no singular solution.

Method of Undetermined Coefficients:

For a linear nonhomogeneous ODE with constant coefficients

$$y^{(n)} + p_{n-1}\, y^{(n-1)} + \cdots + p_1 y' + p_0 y = r(x)$$

we can use the method of Undetermined coefficients in the same way as in second order ODES.

Ex: $y''' - 2y'' - 9y' + 18y = e^{2x}$; $y(0) = 6$, $y'(0) = 9.8$, $y''(0) = 43.2$

The characteristic equation of the associated homogeneous equation : $\lambda^3 - 2\lambda^2 - 9\lambda + 18 = 0$

$$\Rightarrow \lambda^2(\lambda-2) - 9(\lambda-2) = 0 \Rightarrow (\lambda-2)(\lambda^2-9)=0 \Rightarrow (\lambda-2)(\lambda-3)(\lambda+3)=0$$

$$\Rightarrow \lambda = -3,\ 2,\ 3$$

Hence $\quad y_h = c_1 e^{-3x} + c_2 e^{2x} + c_3 e^{3x}$

Now note that the R.H.S. $r(x) = e^{2x}$ and it also satisfies the homogeneous ODE and it corresponds to a single root of the characteristic equation.

Let us try $\quad y_p = A x e^{2x}$

$$y_p' = Ae^{2x} + 2Axe^{2x}$$

$$y_p'' = 2Ae^{2x} + 2Ae^{2x} + 4Axe^{2x} = 4Ae^{2x} + 4Axe^{2x}$$

$$y_p''' = 8Ae^{2x} + 4Ae^{2x} + 8Axe^{2x}$$

$$= 12Ae^{2x} + 8Axe^{2x}$$

Now substituting in the differential equation:

$$12Ae^{2x} + 8Axe^{2x} - 2(4Ae^{2x} + 4Axe^{2x})$$

$$-9(Ae^{2x} + 2Axe^{2x}) + 18Axe^{2x} = e^{2x}$$

$$\Rightarrow 12Ae^{2x} \cancel{+8Axe^{2x}} - 8Ae^{2x} \cancel{-8Axe^{2x}} - 9Ae^{2x}$$ $-x$

$$\cancel{-18Axe^{2x}} + \cancel{18Axe^{2x}} = e^{2x}$$

$$\Rightarrow \quad -5Ae^{2x} = e^{2x} \quad \Rightarrow \quad \boxed{A = -\frac{1}{5} = -.2}$$

So, $\quad y_p(x) = -.2xe^{2x}$

Therefore the general solution is

$$y(x) = c_1 e^{-3x} + c_2 e^{2x} + c_3 e^{3x} - .2xe^{2x}$$

$$y'(x) = -3c_1 e^{-3x} + 2c_2 e^{2x} + 3c_3 e^{3x} - .2e^{2x} - .4xe^{2x}$$

$$y''(x) = 9c_1 e^{-3x} + 4c_2 e^{2x} + 9c_3 e^{3x} - .4e^{2x} - .4e^{2x} - .8xe^{2x}$$

is:

$$= 9c_1 e^{-3x} + 4c_2 e^{2x} + 9c_3 e^{3x} - .8e^{2x} - .8xe^{2x}$$

Now $y(0) = 6 \Rightarrow c_1 + c_2 + c_3 = 6$ ..... ①

$y'(0) = 9.8 \Rightarrow -3c_1 + 2c_2 + 3c_3 - .2 = 9.8 \Rightarrow -3c_1 + 2c_2 + 3c_3 = 10$ ..... ②

$y''(0) = 43.2 \Rightarrow 9c_1 + 4c_2 + 9c_3 - .8 = \quad \Rightarrow 9c_1 + 4c_2 + 9c_3 = 44$ ..... ③

② × 3 + ③ $\Rightarrow 10c_2 + 18c_3 = 74 \Rightarrow 5c_2 + 9c_3 = 37$

① × 3 + ② $\Rightarrow 5c_2 + 6c_3 = 28 \Rightarrow 5c_2 + 6c_3 = 28$

Subtracting $3c_3 = 9 \Rightarrow \boxed{c_3 = 3}$

Then $5c_3 + 27 = 37 \Rightarrow 5c_2 = 10 \Rightarrow \boxed{c_2 = 2}$

Now ① $\Rightarrow c_1 + 2 + 3 = 6 \Rightarrow \boxed{c_1 = 1}$

So, the solution is:

$$\boxed{y(x) = e^{-3x} + 2e^{2x} + 3e^{3x} - .2xe^{2x}}$$

Method Of Variation Of Parameters:

The method of variation of parameters also extends to arbitrary order n.

Let $\{y_1, y_2, \dots, y_n\}$ be a basis of solutions of the homogeneous equation $y^{(n)} + p_{n-1}(x)\, y^{(n-1)} + \dots + p_1(x)\, y' + p_0(x)\, y = 0$

Then a particular solution y_p of the nonhomogeneous equation: $y^{(n)} + p_{n-1}(x)\, y^{(n-1)} + \dots + p_1(x)\, y' + p_0(x) = r(x)$

is given by the formula:

$$y_p(x) = y_1(x) \int \frac{W_1(x)}{W(x)}\, r(x)\, dx + \dots + y_n(x) \int \frac{W_n(x)}{W(x)}\, r(x)\, dx$$

$$= \sum_{k=1}^{n} y_k(x) \int \frac{W_k(x)}{W(x)}\, r(x)\, dx$$

on an open interval I on which $p_{n-1}(x), \ldots, p_1(x), r(x)$ are continuous. Here W is the wronskian of $\{y_1, y_2, \ldots, y_n\}$ and W_k is obtained from W by replacing kth column of W by $\begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix}$

Ex: $\quad x^3 y''' - 3x^2 y'' + 6xy' - 6y = x^4 \ln x \quad (x > 0)$

This is an Euler-Cauchy equation of order 3.
For the general solution of the homogeneous equation we try $y = x^m$.

Then $x^3 m(m-1)(m-2)x^{m-3} - 3x^2 m(m-1)x^{m-2} + 6xmx^{m-1}$
$$-6x^m = 0$$

$$\Rightarrow \quad [m(m-1)(m-2) - 3m(m-1) + 6m - 6]x^m = 0$$

$$\Rightarrow \quad (m-1)[m(m-2) - 3m + 6]x^m = 0$$

$$\Rightarrow \quad (m-1)(m^2 - 5m + 6) = 0 \Rightarrow (m-1)(m-2)(m-3) = 0$$

$$\Rightarrow \quad m = 1, 2, 3$$

$\Rightarrow$ general solution of the homogeneous equation
$$y_h = c_1 x + c_2 x^2 + c_3 x^3$$

Thus $y_1 = x, \quad y_2 = x^2, \quad y_3 = x^3, \quad r(x) = \dfrac{x^4 \ln x}{x^3} = x \ln x$

Now $W = \begin{vmatrix} x & x^2 & x^3 \\ 1 & 2x & 3x^2 \\ 0 & 2 & 6x \end{vmatrix} = x\left(12x^2-6x^2\right) - 1\left(6x^3-2x^3\right)$

$= 6x^3 - 4x^3 = \boxed{2x^3}$

$$W_1 = \begin{vmatrix} 0 & x^2 & x^3 \\ 0 & 2x & 3x^2 \\ 1 & 2 & 6x \end{vmatrix} = 1\left(3x^4 - 2x^4\right) = \boxed{x^4}$$

$$W_2 = \begin{vmatrix} x & 0 & x^3 \\ 1 & 0 & 3x^2 \\ 0 & 1 & 6x \end{vmatrix} = (-1)\left(3x^3 - x^3\right) = \boxed{-2x^3}$$

$$W_3 = \begin{vmatrix} x & x^2 & 0 \\ 1 & 2x & 0 \\ 0 & 2 & 1 \end{vmatrix} = (1)\left(2x^2 - x^2\right) = \boxed{x^2}$$

Then $y_p(x) = x\int \frac{x^4}{2x^3}\, x\ln x\, dx + x^2 \int \frac{(-2x^3)}{2x^3}\, x \ln x\, dx$

$+ x^3 \int \frac{x^2}{2x^3}\, x\ln x\, dx$

$= \frac{1}{2} x \int x^2 \ln x\, dx - x^2 \int x \ln x\, dx + \frac{x^3}{2}\int \ln x\, dx$

$$= \frac{1}{2}x\left[\frac{x^3}{3}\ln x - \frac{x^3}{9}\right] - x^2\left[\frac{x^2}{2}\ln x - \frac{x^2}{4}\right] + \frac{x^3}{2}\left[x\ln x - x\right]$$

$$= \frac{1}{6}x^4\ln x - \frac{1}{2}x^4\ln x + \frac{1}{2}x^4\ln x - \frac{x^4}{18} + \frac{x^4}{4} - \frac{x^4}{2}$$

$$= \frac{1}{6}x^4\ln x - \frac{2x^4 - 9x^4 + 18x^4}{36} = \frac{1}{6}x^4\ln x - \frac{11}{36}x^4$$

$$= \boxed{\frac{1}{6}x^4\left(\ln x - \frac{11}{6}\right)}$$

Thus the general solution of the given ODE is:

$$y(x) = y_h(x) + y_p(x) = \boxed{c_1 x + c_2 x^2 + c_3 x^3 + \frac{x^4}{6}\left(\ln x - \frac{11}{6}\right)}$$

An Example:

$$y'' - 3y' + 2y = 10\sin x$$

For the solution of the homogeneous equation, the characteristic equation is: $\lambda^2 - 3\lambda + 2 = 0$

$$\Rightarrow (\lambda-2)(\lambda-1) = 0 \Rightarrow \lambda = 1, 2$$

So, the solution of the associated homogeneous equation is: $y_h = c_1 e^{x} + c_2 e^{2x}$

- For a particular solution of the nonhomogeneous equation, first we use Method of Undetermined coefficients.

Let $y_p = A\sin x + B\cos x$

Then $y_p' = A\cos x - B\sin x$ and $y_p'' = -A\sin x - B\cos x$

Then $[-A\sin x - B\cos x] - 3(A\cos x - B\sin x) + 2[A\sin x + B\cos x]$
$= 10\sin x$

$\Rightarrow \sin x[-A + 3B + 2A] + \cos x[-B - 3A + 2B] = 10\sin x$

$\Rightarrow \left.\begin{array}{r} A + 3B = 10 \\ -3A + B = 0 \end{array}\right\} \Rightarrow 10A = 10 \Rightarrow A = \boxed{1}$
and $B = \boxed{3}$

So, $\boxed{y_p(x) = \sin x + 3\cos x}$

We can also use method of Variation of Parameters to obtain $y_p(x)$:

$y_1 = e^x, \quad y_2 = e^{2x}$

$$W = \begin{vmatrix} e^x & e^{2x} \\ e^x & 2e^{2x} \end{vmatrix} = 2e^{3x} - e^{3x} = e^{3x}$$

Then $u(x) = -\int \frac{e^{2x}}{e^{3x}} 10\sin x \, dx = -10\int e^{-x}\sin x \, dx$

$= -10\left[\sin x(-e^{-x}) + \int \cos x \, e^{-x} dx\right]$

$$= 10e^{-x}\sin x - 10\left[\left(-e^{-x}\cos x\right) - \int \sin x\, e^{-x}\,dx\right]$$

$$= 10e^{-x}\sin x + 10e^{-x}\cos x - u(x)$$

$$\Rightarrow 2u(x) = 10e^{-x}\left[\sin x + \cos x\right]$$

$$\Rightarrow u(x) = 5e^{-x}\left[\sin x + \cos x\right]$$

Now $v(x) = \displaystyle\int \frac{e^{x}}{e^{3x}} 10\sin x\,dx = 10\int e^{-2x}\sin x\,dx$

$$= 10\left[\frac{e^{-2x}}{(-2)}\sin x - \int \frac{e^{-2x}}{(-2)}\cos x\,dx\right]$$

$$= 10\left[-\frac{1}{2}e^{-2x}\sin x + \frac{1}{2}\left\{\frac{e^{-2x}}{(-2)}\cos x - \int \frac{e^{-2x}}{(-2)}(-\sin x)\,dx\right\}\right]$$

$$= -5e^{-2x}\sin x - \frac{5}{2}e^{-2x}\cos x - \frac{5}{2}\int e^{-2x}\sin x\,dx$$

$$= -5e^{-2x}\left[\sin x + \frac{1}{2}\cos x\right] - \frac{1}{4}v(x)$$

$$\Rightarrow \frac{5}{4}v(x) = -5e^{-2x}\left[\sin x + \frac{1}{2}\cos x\right]$$

$$\Rightarrow v(x) = -4e^{-2x}\sin x - 2e^{-2x}\cos x$$

So, $y_p(x) = u(x)y_1(x) + v(x)y_2(x)$

$$= e^{-x}\left[5e^{-x}(\sin x + \cos x)\right] + e^{-2x}\left[-4e^{-2x}\sin x - 2e^{-2x}\cos x\right]$$

$$= 5\sin x + 5\cos x - 4\sin x - 2\cos x$$

$$\Rightarrow \boxed{y_p(x) = \sin x + 3\cos x}$$